Sara Alrifai

Basc. Mechatronics Engineering | University of Waterloo

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Skills

Software/Programming: C, C++, Matlab, VHDL (FPGA), RobotC.
Electronics & Embedded Systems: PLC programming, Arduino, STM32, FPGA development, PCB & circuit design, Micro soldering, Sensors/Signals.
CAD/Prototyping: AutoCAD, SolidWorks, GD&T, Freehand sketching, FDM 3D printing, Tool usage, Hardware building.
Communication: Technical writing, Fluent in English & Arabic, Intermediate in Turkish & French.

Experience

Administrative Support Assistant Jan - Apr 2025
University of Waterloo, Engineering Undergraduate Office *Waterloo, ON*

- Performed advanced Excel analysis (PivotTables, VLOOKUP) to support planning and reporting.
- Streamlined digital records for rapid, reliable data retrieval.
- Provided efficient front-desk service, assisting 30+ daily users with professional communication.
- Managed Iron Ring Ceremony logistics using structured workflows and version-controlled documentation.

CAD Modelling Assistant Apr - Sep 2023
Business Teams *Mississauga, ON*

- Produced detailed floor plans and 3D models using **AutoCAD** and **SolidWorks**.
- Collaborated with engineers to meet design specifications and client requirements.
- Created precise **technical drawings** showcasing strong attention to detail.

Project Team Member, Digital Bootcamp May - Aug 2023
WE Accelerate Program *Remote*

- Led UX-focused website redesign using Agile methods to improve usability.
- Implemented new features with CMS tools ensuring responsive, accessible layouts.
- Conducted user testing and analyzed feedback to inform design improvements.
- Coordinated virtual meetings, delegated tasks, and delivered milestones on time.

Volunteer Teacher 2022 - Present
Al Siraj Al Munir

- Managed student behavior and developed new learning techniques.

Volunteer Event Coordinator 2019 - 2021
MYNA

- Organized volunteer opportunities and food drives across Ontario.

Projects

Robo Prosthetic Hand 2023

- Designed and programmed a functional prosthetic hand using **C**, capable of controlled object handling to act as a replacement for human hand.

Arduino Speaker 2023

- Built and programmed a tone-generating speaker system using **Arduino**, to modulate soundbased on input signals.

Education

University of Waterloo, BASc. Mechatronics Engineering 2023 - Exp. 2028
Relevant Coursework: Microprocessors and Digital Logic, Algorithms and Data Structures (C++/RobotC), Circuits, Materials Engineering, Mechanics of Deformable Solids, Statics & Dynamics, Statistical Analysis, Sensors and Instrumentation, Real-Time Systems, MATLAB.

Certifications

WHMIS (2023), CPR/First Aid (C), G Driver's License